PillPlan

Technical Documentation

Architecture, design system, data, testing, performance and deployment.

Version	2.0, 2 November 2026
Stack	React 19, TypeScript 6, Vite 8, Python 3.12 with Pydantic 2
Hosting	Netlify, static files only
Live	`https://pillplan-cela.netlify.app`
Repository	`https://github.com/4arkeinlvy/pillplan`

> **At a glance.** 43 medicines, 132 automated tests, 93.6 KB of initial JavaScript, no server and no database. Everything the user types stays in their own browser.

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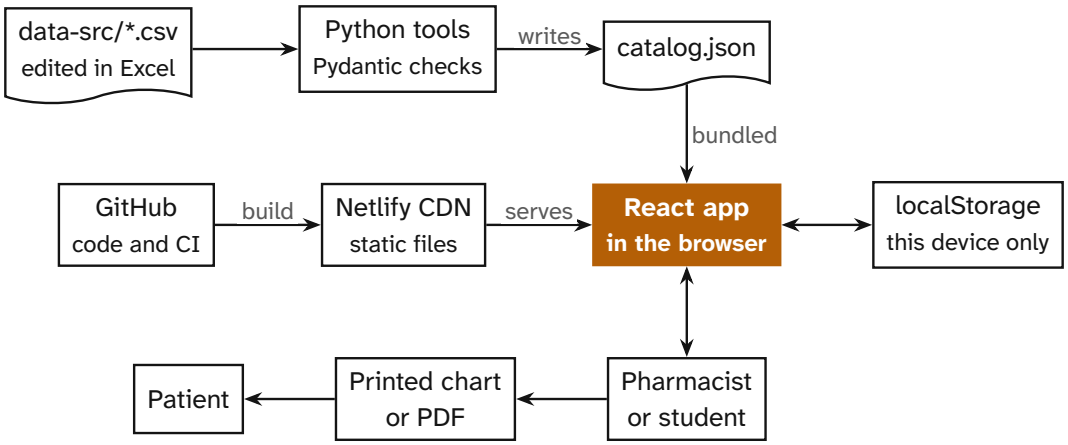
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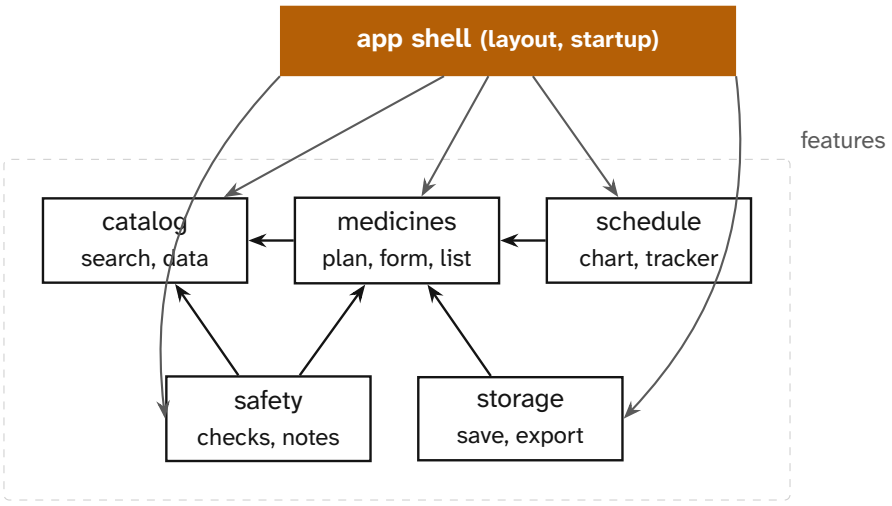
1 System context

PillPlan is a static website. Python runs only on the developer machine and in CI, where it checks the drug data and writes the JSON file the app loads. In the browser, the React app does all the work.



2 Architecture: a modular monolith

The app is **one codebase, one build and one deploy**, split into five feature modules with strict borders. Each module has a public `index.ts`. Other code may import only from that file. A shared kernel holds what every module agrees on.



shared kernel: intervals and clock times, dose format, strings, icons, UI primitives, tokens

Black arrows are allowed dependencies between features. They always point one way, so there are no cycles.

2.1 Rules the linter enforces

Rule	ESLint check
Use a feature through its door	`@/features/x/inner/file` is refused. Only `@/features/x` is allowed.
No reaching across with paths	`../../other-feature` is refused inside features.
Shared stays generic	`shared` may not import any feature or the app shell.
No cycles	`import-x/no-cycle`.
No raw HTML injection	`react/no-danger`.

2.2 Shared kernel: two languages

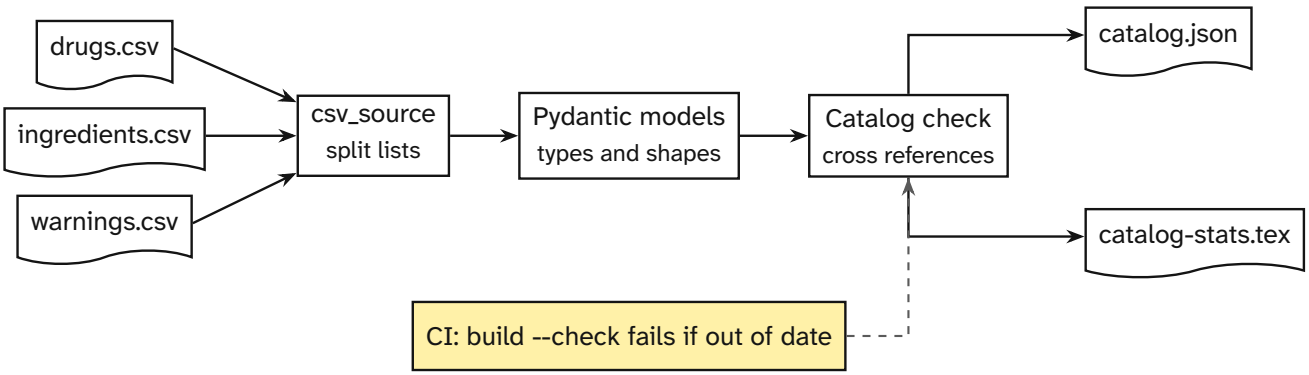
`shared/i18n` holds one dictionary per language, `en.ts` and `id.ts`. The Indonesian one is typed as `Strings`, so TypeScript refuses a missing or misspelled key. Logic functions take the dictionary as a parameter (`validateMedicine(m, t)`, `runSafetyChecks(meds, catalog, t, lang)`); components read it with `useI18n()`. Catalog text such as advice and warnings is stored as `{en, id}` pairs. Numbers follow the locale: `0.5 mg` and `4,000 mg` in English, `0,5 mg` and `4.000 mg` in Indonesian, always with a leading zero and no trailing zero.

2.3 Inside a feature

Every feature uses the same layout:

- `model.ts`: types.
- `logic/`: pure TypeScript functions with no React, each with tests.
- `ui/`: thin React components, each with its own CSS Module file. Class names are scoped to that component and Vite merges all styles into one small CSS file.
- `index.ts`: the public API.

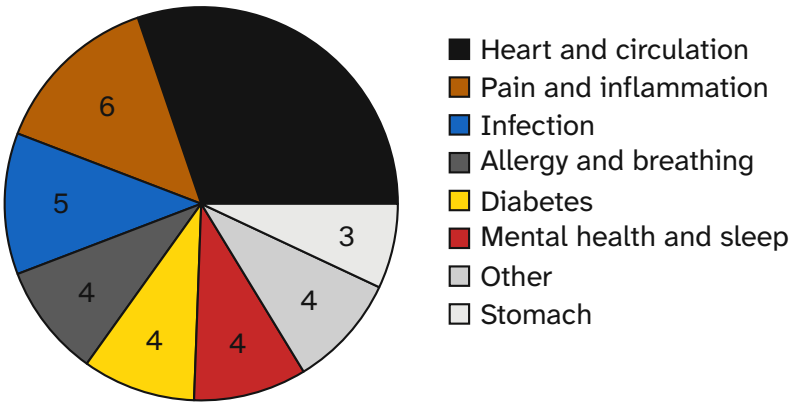
3 Data pipeline



The catalog check refuses:

- a duplicate id, an unknown ingredient or an unknown warning;
- a strength that does not give one amount per ingredient;
- a unit that does not match its ingredient;
- a single tablet stronger than the usual daily maximum;
- any em or en dash in the text.

3.1 What is in the catalog



43 medicines in 34 drug classes and 8 areas. 35 of 43 ingredients have a usual daily maximum, 9 medicines have a best time of day, and there are 9 warning stickers.

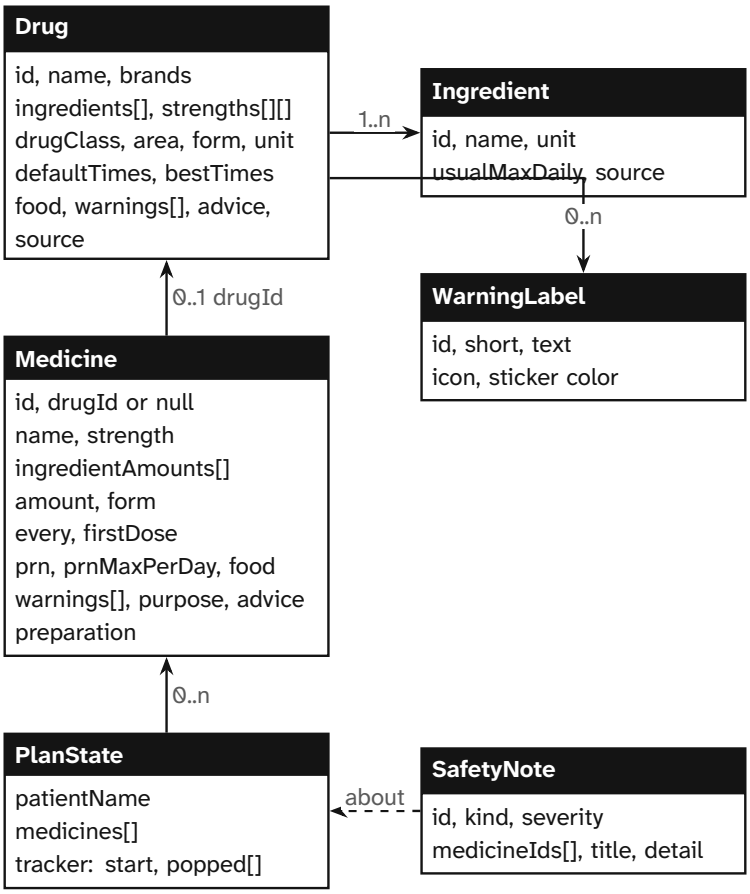
4 Signa entry

`parseSigna()` in `features/medicines/logic/signa.ts` is a pure function with 34 tests. It reads Latin and Indonesian aturan pakai and returns an interval and first dose, amount, form, food, “when needed” and a list of words it did not understand, which the form shows as “Check: …”.

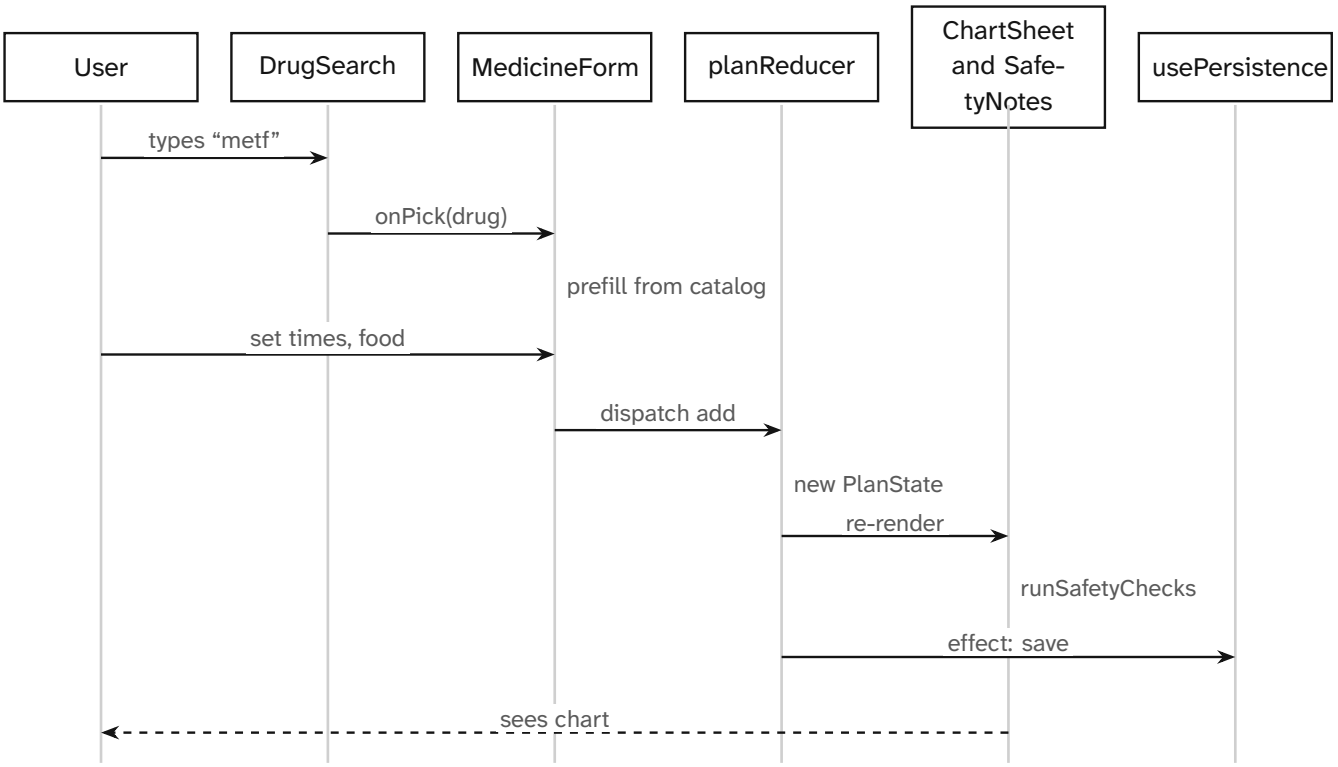
Signa	Meaning in PillPlan
`1 dd, 1x, o.d., q.d.`	every 24 hours, at the drug's usual time or 07.00
`2 dd, 2x, b.i.d., tiap 12 jam`	every 12 hours: 07.00 and 19.00
`3 dd, 3x, t.i.d., tiap 8 jam, q8h`	every 8 hours: 07.00, 15.00 and 23.00
`4 dd, 4x, q.i.d., tiap 6 jam`	every 6 hours: 07.00, 13.00, 19.00 and 01.00
`o.m., pagi`	first dose at 07.00 (omni mane)
`o.n., malam`	first dose at 19.00 (omni nocte)
`h.s., sebelum tidur`	first dose at 21.00 (hora somni)
`a.c., sebelum makan`	before food (ante coenam)
`p.c., d.c., sesudah/saat makan`	with or after food (post or durante coenam)
`p.r.n., s.o.s., bila perlu, k/p`	when needed; "3 dd 1 prn" means at most 3 a day
`1, ½, 1/2, 0,5, I, II, s`	amount; s (semis) is ½
`tab, caps, cth, C, gtt`	tablet, capsule, 5 mL, 15 mL, drops

More than 4 doses a day (for example `tiap 4 jam`) is never guessed: the parser flags it for the apoteker.

5 Data model

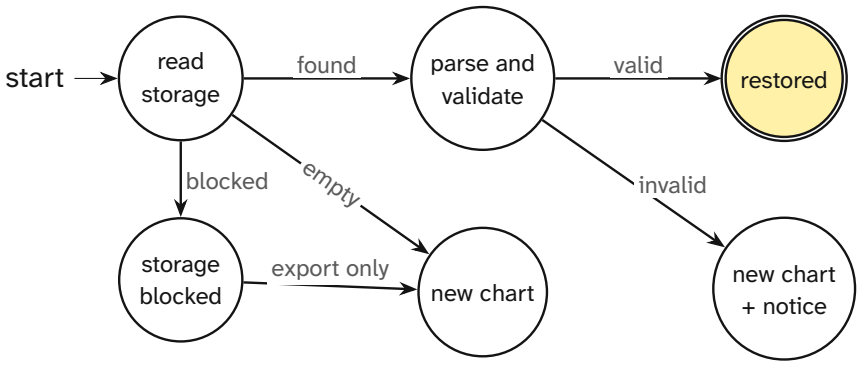


6 Adding a medicine



The add is wrapped in a View Transition, so the new row slides in. Browsers without View Transitions simply update.

7 Saving and loading



The browser copy and an exported file use the same envelope: `{app, version, savedAt, plan}`. Valibot checks it before use. The first dose must be a valid `HH:MM` and the interval one of 24, 12, 8 or 6 hours. Files from version 1, which used morning, noon, evening and bedtime, are migrated on load: the number of times becomes the interval (three times is every 8 hours from 07.00) and the pill shape becomes a sediaan.

8 Phone reminders

`buildIcs()` writes an iCalendar (RFC 5545) file in the browser: one daily event per time of day in use, each with an alarm, repeating for 30 days, in floating local time so it rings at the same clock

time wherever the phone is. Text is escaped and long lines are folded at 75 bytes. The default times are 07:00, 12:00, 19:00 and 21:00 and can be changed before download. When-needed medicines get no reminder.

9 Design system

9.1 Direction: a dispensing label

The look comes from real pharmacy objects, not from a template:

- white label paper, black ink and ruled lines;
- one amber accent, the color of prescription bottles;
- warning stickers in the colors of auxiliary labels;
- a blister pack for the tracker.

9.2 Color tokens

Token	Hex	Use	Contrast on white
`ink`	`#141414`	Text and rules	18.4 : 1
`ink-muted`	`#5A5A5A`	Secondary text	6.9 : 1
`rule`	`#CFCFCF`	Lines	decorative
`amber`	`#B45F06`	Primary buttons, focus ring	4.6 : 1
`alert`	`#C62828`	Danger, red stickers	5.6 : 1
`sticker-yellow`	`#FFD60A`	Caution stickers, black text	14.0 : 1 (ink)
`sticker-blue`	`#1565C0`	Information stickers	5.8 : 1
`highlight`	`#FFF1A8`	Spotlight, active option	background

Stickers always combine color, an icon and words: **May cause sleepiness** **No alcohol** **Swallow whole**

9.3 Typography

- **Atkinson Hyperlegible Next** for all text. The Braille Institute designed it so letters that are easy to confuse look different.
- `Atkinson Hyperlegible Mono` for dose numbers, where 0, O, 1, l and I must never be confused.
- Self-hosted woff2 files, latin only, weights 400 and 700, preloaded.
- 18 px base and a 1.25 scale. Print uses 14 pt for names and doses.

9.4 Design rules

These rules keep the screen calm for a busy counter. Every new screen is checked against them.

Rule	In PillPlan
One task per area	Left: build the plan. Right: the chart. Everything else (save, PDF, reminders, my drugs) is in the More drawer. Safety notes are one line above the chart until opened.
One primary button	Amber only for the main action of a view: Print / PDF, Add to chart, Add to phone calendar.
Light controls, strong content	Inputs, chips and buttons have 1 px light grey borders that darken on hover and focus. Strong black lines appear only on the chart sheet.
Ruled grid	The desk is a faint cutting mat: lines every 8 px and stronger ones every 40 px. The A4 chart lies on it like label paper. The layout uses 12 columns with a 24 px gutter: the plan takes 4, the chart 8.
Navbar	No boxed buttons. Each icon answers its meaning on hover: the printer feeds paper, the speech bubble pops, the three dots spread. A 2 px line grows under the label. Print / PDF is the only filled button.
Tooltips	Only for short help, never for anything needed to finish the task. They open on hover after 400 ms and at once on keyboard focus, close with Esc, stay open while the pointer is on them, and never show on touch screens (WCAG 2.2 SC 1.4.13).
No stripe accents	Severity is shown by an icon and the title colour, never by a coloured left border.
8 px spacing grid	32 px between sections, 16 px between fields, 6 px from label to control.
Type scale	17 px interface text, 18 px chart text, headings 20, 24 and 32 px. Bold only for names, doses and labels.
Touch targets	Every control is at least 44 px tall (WCAG 2.2).
Colour budget	About 60% paper and desk, 30% ink, at most 10% amber. A chosen chip is an amber tint, not solid black.
Alignment	Fields in a row share one top line. Units sit inside the stepper (“1 tablet”, “4 kali”), never below it.
Motion	Material 3 tokens: standard easing `(0.2, 0, 0, 1)`, enter `(0.05, 0.7, 0.1, 1)`, exit `(0.3, 0, 0.8, 0.15)`; 150 ms for feedback, 300 ms for state changes, 450 ms for the one orchestrated moment. No bounce, no loops, off with reduced motion.

9.5 Icons

Dose times are written as clock times (07.00, 15.00, 23.00). Each one gets a small sun or moon picked from the hour, only as a visual aid: a rising sun before 11.00, a full sun until 15.00, a moon with a star until 20.00 and a bed with “z” at night. The sediaan has its own drawings: a scored tablet, a two-tone capsule, a syrup bottle, and a suspension bottle with sediment and shake marks.

The icons are drawn as 24 px SVG with 2 px strokes, so they print sharply.

9.6 Anti-slop rules

Taken from Anthropic's `frontend-design` skill and the `avoid-ai-design` catalog. PillPlan does not use:

- Inter, Roboto or system-ui as the design font;
- purple or blue gradients;
- the cream, terracotta and serif look;
- identical rounded cards with the same soft shadow;
- tracked capital labels or one highlighted word in a headline;
- fade-up on every section;
- sparkle icons or icons in rounded squares;
- marketing copy.

Motion is used only where it explains a change, built with CSS and the View Transitions API, so the app's main bundle adds no animation library:

Moment	What moves and why
Chart printing	The one orchestrated moment: the chart appears like a label printer printing, each row revealed left to right, one after another.
Dose stamp	Then each dose is pressed into its column like a pharmacy stamp, column by column. It shows where the dose landed.
Navbar icons	Hover or focus makes each icon act out its meaning (paper feed, bubble, dots).
Signa preview	The preview turns bold, a check mark draws itself, and the time chips switch on with their icons moving.
Drawer and strip	The More drawer slides in from the right; the safety strip opens downward with a turning chevron.
Dose clock	The doses sit on a 24-hour dial joined into one shape. Dragging the first dose turns the shape along the ring, so the gaps stay equal.
Welcome to app	Leaving the welcome film, the PillPlan mark glides into the navbar while the rest cross-fades, in one 450 ms view transition.
Counseling page turn	Next slides the medicine to the left, Back to the right, so the direction is clear.
Note to rows	Pointing at a safety note outlines the medicines it is about and fades the rest.
Day ribbon	The bars grow and shrink as medicines are added or removed.
Rows and language	Rows slide when added, removed or reordered; the language switch cross-fades.
Tracker and notes	A blister bubble pops when ticked; a new safety note arrives like a slapped sticker.

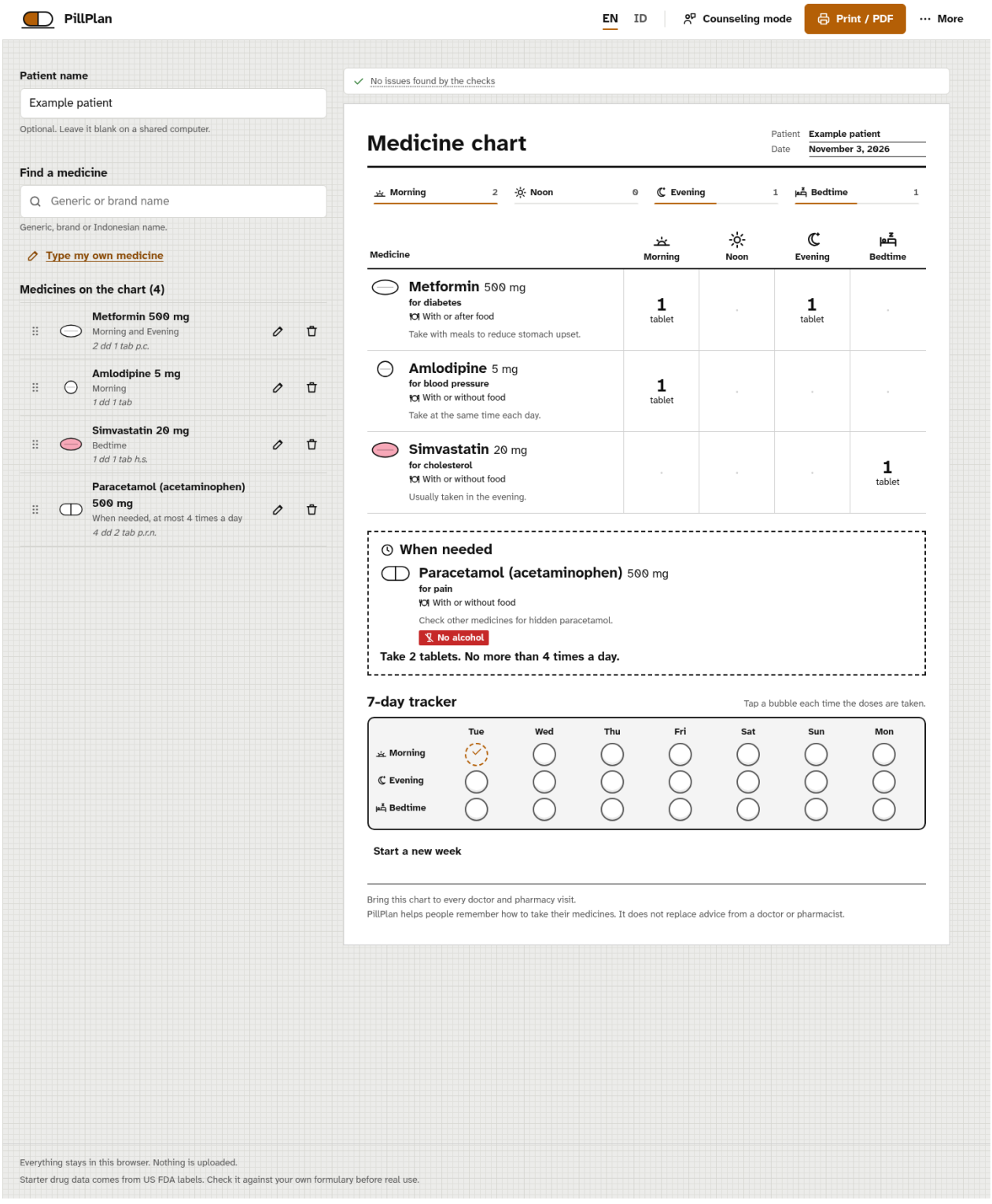
9.7 Welcome film and 3D model

The welcome film is a Remotion project in `web/video/` that is *rendered ahead of time*, not played live. `npm run video:render` writes, for each language, a VP9 webm with a transparent background (the page grid shows through it), an H.264 mp4 on the desk colour for Safari, and a poster image. Each file must stay under 600 KB or the script stops. The site plays them with a plain `<video>`, muted, with a sound toggle, so no Remotion code reaches the browser. The English film has a narration from the public domain LJSpeech voice (Piper TTS, offline, `npm run video:voice`); both films have generated sound effects. Rendering uses the GPU through ANGLE when it is available.

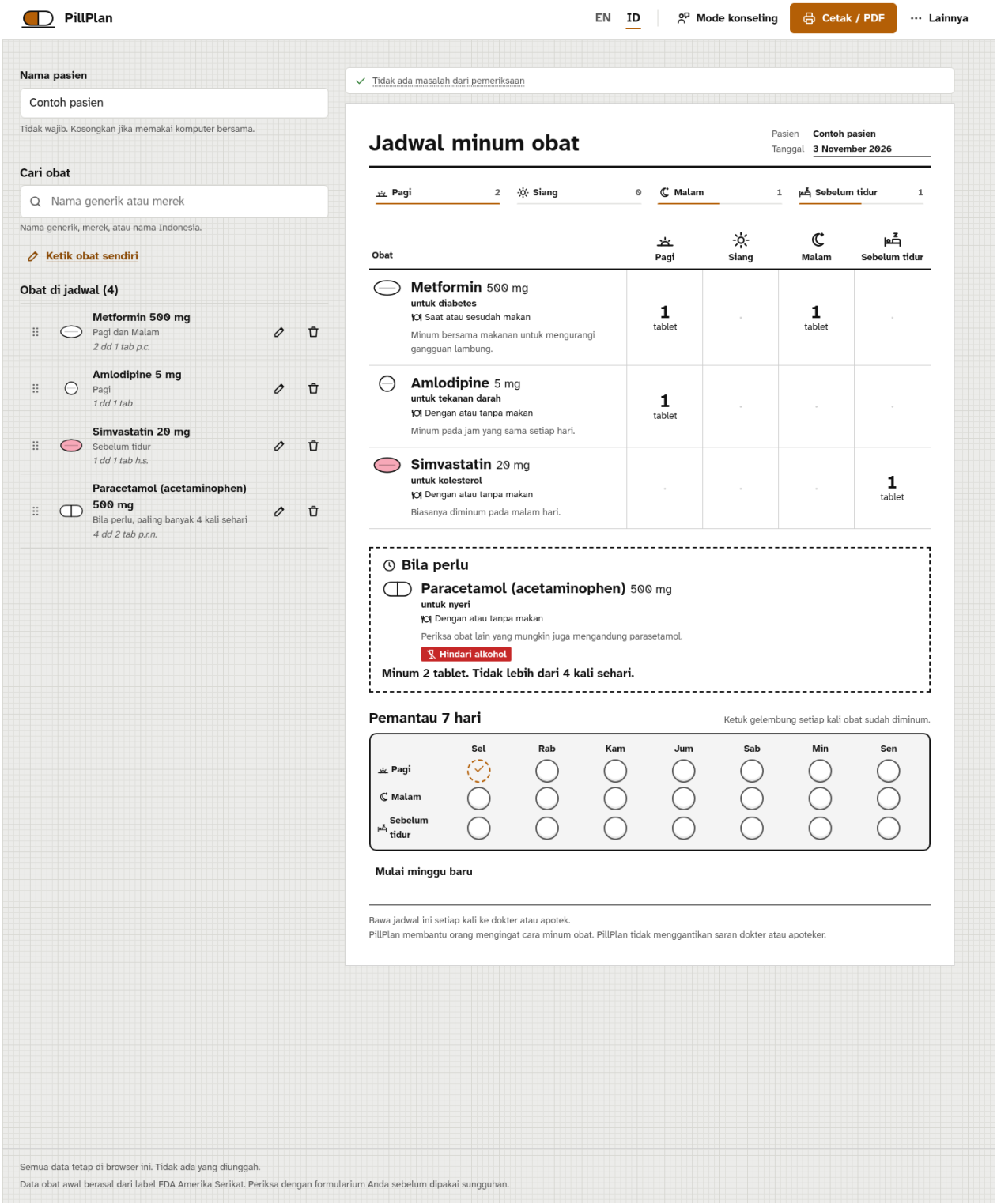
Counseling mode shows the sediaan as a 3D model built from simple shapes with three.js and react-three-fiber. It is loaded only there, renders only when it is dragged, and falls back to the flat drawing without WebGL or with reduced motion.

Same-document View Transitions work in Chrome, Edge, Firefox 144+ and Safari 18+; other browsers simply update. All motion is switched off when the device asks for reduced motion.

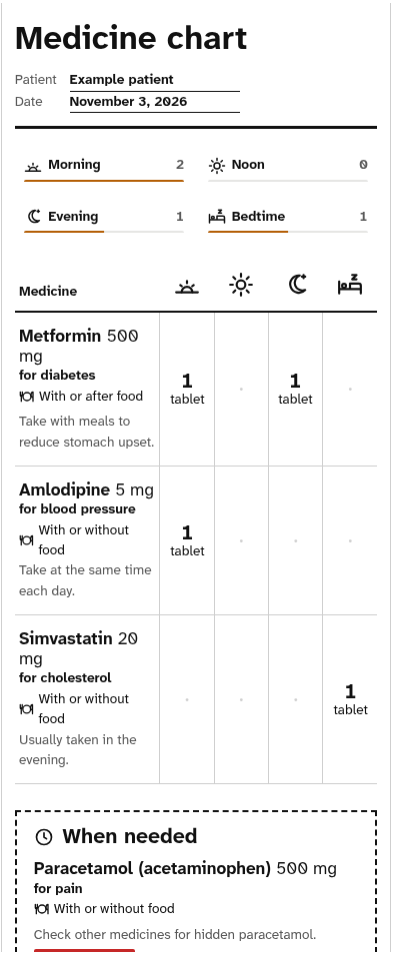
9.8 Screens



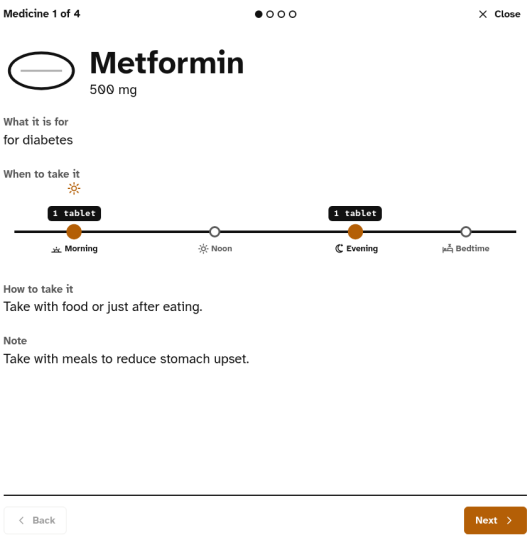
Desktop, 1440 px: form and list on the left, the A4 chart on the right.



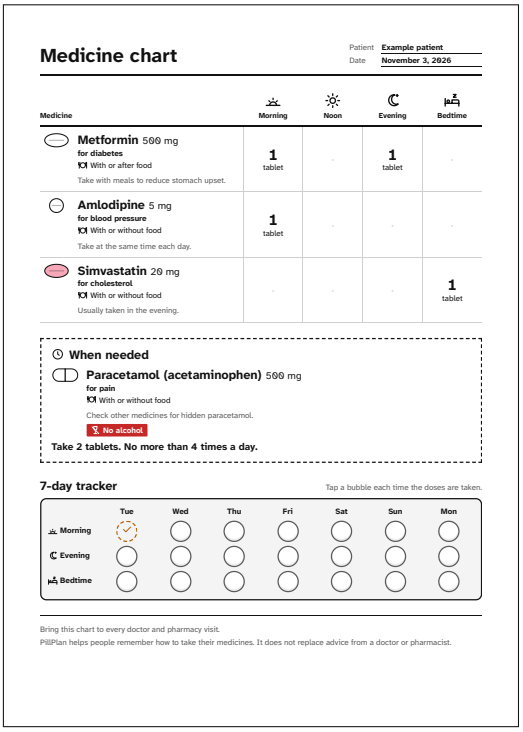
The same plan in Indonesian.



Phone, 390 px



Counseling mode



Print, A4

10 Safety checks

Check	Level	How
Daily maximum	alert	For each ingredient: amount per unit × units per dose × doses per day, summed across every product, compared with the usual maximum.
Same ingredient	caution	Two or more products share an ingredient, for example paracetamol in Panadol and Panadol Extra.
Same class	caution	Two or more drugs from one specific class, for example two statins. Aspirin and clopidogrel are different classes.
Time of day	info	A medicine set outside its best times, for example simvastatin in the morning.
Not checked	info	A custom medicine that is not in the catalog.

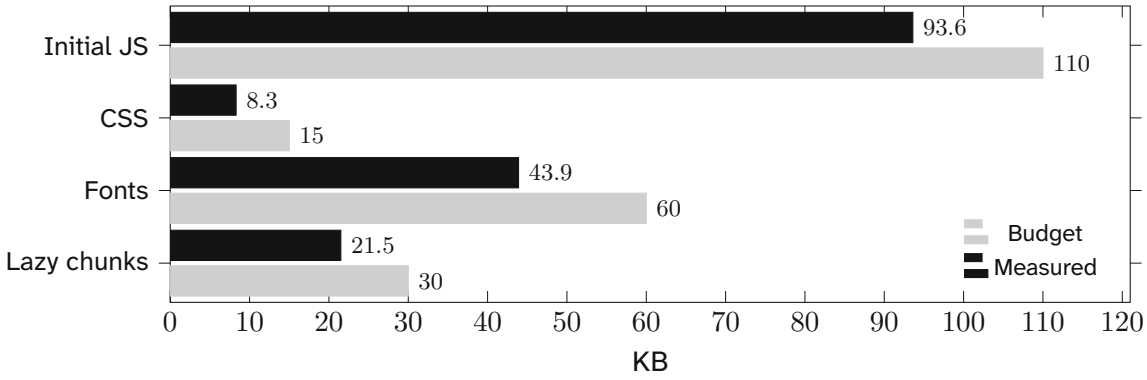
11 Testing

Suite	Tests	Covers
Web (Vitest)	117	Dose formatting, search, reducer, prefill, validation, chart, tracker, safety checks, storage and files, and full user flows in jsdom
Data (pytest)	15	Real catalog validity and every refusal rule
Total	132	all passing

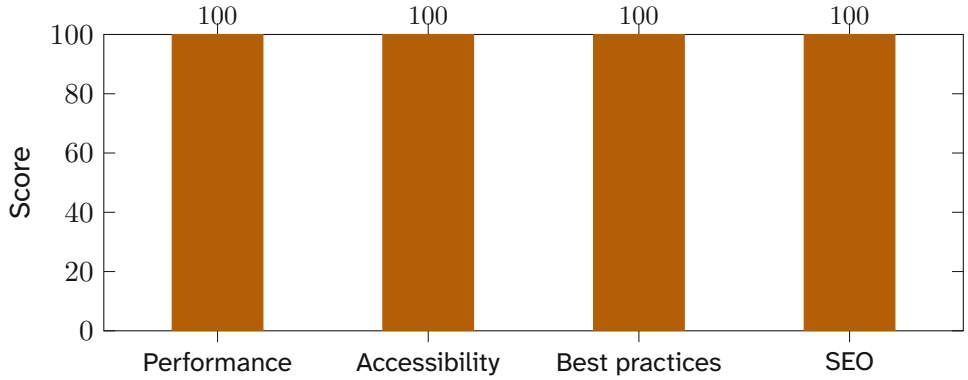
The user-flow tests use Testing Library and query by role and accessible name, the same way a screen reader user finds things:

- add from search;
- hidden paracetamol and undo;
- custom medicine;
- validation;
- counseling mode with arrow keys;
- tracker saving;
- spotlight;
- turning saving off.

12 Performance



JavaScript and CSS sizes are gzip. Initial JavaScript is 81.4 KB with Brotli, which Netlify serves. The drug list and counseling mode are separate chunks loaded when needed.



Lighthouse, mobile: local production build, mobile, LCP 0.8 s, blocking time 0 ms, layout shift 0.

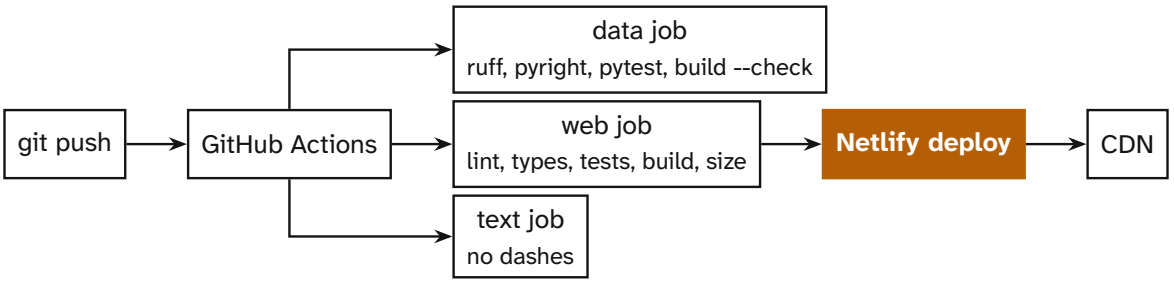
13 Security

- No server, database, accounts, cookies, analytics or third-party scripts.
- Headers from `web/public/_headers` on every response:

```
Content-Security-Policy: default-src 'self'; img-src 'self' data:;
  object-src 'none'; base-uri 'none'; frame-ancestors 'none'; form-action 'none'
Referrer-Policy: no-referrer
X-Content-Type-Options: nosniff
Permissions-Policy: camera=(), microphone=(), geolocation=()
Cross-Origin-Opener-Policy: same-origin
```

- React escapes all text, and `dangerouslySetInnerHTML` is banned by lint.
- Saved data and imported files are validated with Valibot and limited in size.

14 Build, CI and deploy



14.1 Deploy step by step

1. `cd web && npm run build`
2. `npx netlify-cli login`, once.
3. `npx netlify-cli deploy --prod --dir web/dist`
4. Optional: in Netlify, choose *Link repository* so every push deploys. The settings come from `netlify.toml`.

15 Editing the drug data

1. Open `data-src/drugs.csv` in Excel or Google Sheets.
2. Lists use semicolons, such as `morning;evening`. Combination strengths use plus, such as `500+65`. Use `prn` for when needed.
3. Every drug needs a `source`.
4. Run `cd tools && uv run python -m pillplan_tools.catalog.build`. Any mistake is reported with the drug id.
5. Run the tests, then commit.

Starter data. Values are based on US FDA labeling and are for education. Check them against your own formulary before any real use.